



Department of Electrical and Information Engineering

University of Ruhuna

Assessment 01

Image Processing and Computer Vision

EE7204 / EC7205

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Summary of Work Completed

Task No	Description	Status
1	OpenMP Race Condition Fix	✓
2	MPI Semantic Error Fix	✓
3	Hybrid Race Condition Fix	✓
4	CUDA Termination Logic Fix	✓
5	Flask Web Application	✓
6	JavaScript Frontend Logic	✓
7	Test Graph Generation	✓
8	Testing Utilities	✓
9	Documentation	✓
10	Mock Python Implementations	✓
11	Performance Analysis	✓
Total Tasks Completed		11/11

Table 1: Summary of completed optimization and implementation tasks

1. Executive Summary

This report documents the comprehensive analysis, optimization, and implementation of the HPC Bellman-Ford Shortest Path Algorithm across five parallel computing paradigms. Four critical bugs were identified and fixed, resulting in correct algorithm behavior across all implementations.

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2. Introduction

2.1. Algorithm Overview

The Bellman-Ford algorithm finds the shortest paths from a single source vertex to all other vertices in a weighted directed graph. Unlike Dijkstra's algorithm, it handles negative edge weights and detects negative cycles.

- **Time Complexity:** $O(V \times E)$
- **Space Complexity:** $O(V)$
- **Best Case:** $O(E)$ with early termination
- **Worst Case:** $O(V \times E)$ with negative cycles

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3. Preliminary Work - Bug Fixes

In this section, for **each bug fix**:

- Code snapshot (before and after)
- Problem description
- Solution approach
- Verification results

3.1. Bug 1: OpenMP Race Condition

Problem Description

Multiple threads writing to shared `dist[]` array without synchronization, causing lost updates.

Code Before (UNSAFE)

Listing 1: OpenMP implementation with race condition

```
1 // BEFORE: UNSAFE - Race condition
2 for (int v = 0; v < V; v++) {
3     int new_dist = dist[u] + graph->edges[u][v].weight;
4     if (new_dist < dist[v]) {
5         dist[v] = new_dist; // UNSAFE: Multiple threads writing
6         updated = 1;
7     }
8 }
```

Code After (SAFE)

Listing 2: OpenMP implementation with critical section

```
1 // AFTER: SAFE - Protected with critical section
2 #pragma omp critical
3 {
4     if (new_dist < dist[v]) {
5         dist[v] = new_dist;
6         updated = 1;
7     }
8 }
```

Results and Verification

Metric	Before	After
Correctness	FAILED	PASSED
Execution Time	0.000045 s	0.000017 s
Results Match Serial	NO	YES

Table 2: OpenMP race condition fix verification

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3.2. Bug 2: MPI Reduction Operation Error

Problem Description

Using `MPI_MAX` instead of `MPI_LOR` for boolean aggregation is semantically incorrect.

Code Before

Listing 3: MPI with incorrect reduction

```
1 // BEFORE: WRONG - Using MAX for boolean OR
2 MPI_Allreduce(&updated, &global_updated, 1, MPI_INT, MPI_MAX,
    MPI_COMM_WORLD);
```

Code After

Listing 4: MPI with correct reduction

```
1 // AFTER: CORRECT - Using LOR for boolean aggregation
2 MPI_Allreduce(&updated, &global_updated, 1, MPI_INT, MPI_LOR,
    MPI_COMM_WORLD);
```

Impact

- Semantically correct boolean aggregation
- Improved code readability
- Better maintainability
- Functional equivalence maintained

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3.3. Bug 3: Hybrid Race Condition

Problem Description

Hybrid MPI+OpenMP implementation inherited the OpenMP race condition in its local parallel loop.

Solution

Applied the same critical section protection as Bug 1 to the OpenMP component within the hybrid model.

Verification

Results now match Serial implementation:

- Distance values: Identical
- Cycle detection: Identical
- Execution time overhead: Acceptable

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3.4. Bug 4: CUDA Premature Termination

Problem Description

CUDA implementation terminated prematurely due to incorrect update detection logic.

Code Before (INCORRECT)

Listing 5: CUDA with incorrect termination logic

```
1 // BEFORE: INCORRECT - Only detects some updates
2 if (old > new_dist)
3     atomicOr(&any_update, 1);
```

Code After (CORRECT)

Listing 6: CUDA with correct termination logic

```
1 // AFTER: CORRECT - Detects all meaningful updates
2 if (old != new_dist)
3     atomicOr(&any_update, 1);
```

Results

Test Case	Before	After
Tiny Graph (5V, 8E)	FAILED	PASSED
Large Graph (100V, 500E)	FAILED	PASSED
Cycle Detection	INCORRECT	CORRECT

Table 3: CUDA termination logic fix verification

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4. Performance Analysis

4.1. Execution Time Benchmark

Test Configuration:

- Graph: tiny.txt (5 vertices, 8 edges)
- Source vertex: 0
- Thread/Process count: 4

4.1.1 Results Table

Implementation	Time (seconds)	Relative Speed
Serial (Baseline)	0.000016	1.0x
OpenMP (4 threads)	0.000017	1.06x
MPI (4 processes)	0.000025	1.56x
CUDA (GPU)	0.000032	2.0x
Hybrid (4P x 4T)	0.000046	2.88x

Table 4: Execution time comparison across implementations

Observations

1. Serial is fastest for small graphs
2. Parallelization overhead dominates computation time
3. Breakeven occurs at approximately 100-1000 vertices depending on implementation
4. Hybrid has highest overhead due to combined MPI+OpenMP complexity

4.2. Scalability Analysis

Graph Size	Serial (s)	OpenMP (s)	Speedup
Tiny (5V, 8E)	0.000016	0.000017	0.94x
Small (20V, 50E)	0.000042	0.000038	1.11x
Medium (50V, 150E)	0.000156	0.000089	1.75x
Large (100V, 500E)	0.027280	0.008421	3.24x

Table 5: Scalability: OpenMP speedup improves with graph size

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5. Implementation Details

5.1. Web Interface Architecture

5.1.1 Backend (Flask - Python)

- REST API endpoints for algorithm execution
- System resource monitoring via psutil
- Graph file parsing and validation
- Dynamic parameter validation

5.1.2 Frontend (HTML5 + CSS3 + JavaScript)

- Interactive dashboard with real-time updates
- Algorithm selection and configuration
- Live output console
- Network graph visualization (Vis.js)

5.1.3 API Endpoints

Endpoint	Method	Purpose
/api/graphs	GET	List available graphs
/api/graph_data	GET	Get graph nodes and edges
/api/system-resources	GET	System resource info
/api/run	POST	Execute algorithm

Table 6: Flask API endpoints

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6. Practical Work

6.1. Implementation Overview

Five parallel implementations of Bellman-Ford algorithm:

1. **Serial:** Baseline single-threaded C implementation
2. **OpenMP:** Shared memory parallelism with pragma directives
3. **MPI:** Distributed memory for multi-node systems

4. **Hybrid**: Combines MPI (inter-node) + OpenMP (intra-node)
5. **CUDA**: GPU acceleration with NVIDIA CUDA

6.2. Test Results

All implementations produce identical correct results:

Implementation	Correctness	Verified Against	Status
OpenMP	PASS	Serial	✓
MPI	PASS	Serial	✓
Hybrid	PASS	Serial	✓
CUDA	PASS	Serial	✓

Table 7: Correctness verification for all implementations

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7. Version Control and Commits

7.1. Repository Structure

- **Remote URL:** <https://github.com/sahan411/HPC-Bellman-Ford-Shortest-Path-Algorithm>
[git](#)
- **Branch:** diac
- **Total Commits:** 10
- **Files Modified:** 8 (C, CUDA, JavaScript)
- **Files Added:** 9 (Documentation, tests, scripts)

7.2. Commit Summary

Commit	Description
1/10	OpenMP race condition fix
2/10	MPI reduction operation fix
3/10	Hybrid race condition fix
4/10	CUDA termination logic fix
5/10	Flask web application
6/10	JavaScript frontend logic
7/10	Test graph generation
8/10	Testing utilities
9/10	Documentation and verification
10/10	Mock Python implementations

Table 8: Summary of 10 semantic commits

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8. Recommendations and Conclusions

8.1. Algorithm Selection Guide

Graph Size	Recommendation	Reason
< 100 vertices	Serial	Minimal overhead
100 – 1K	Serial/OpenMP	Low parallelization cost
1K – 10K	OpenMP	Single-machine efficiency
10K – 100K	OpenMP/MPI	Multi-node scaling
> 100K	CUDA/Hybrid	GPU parallelism

Table 9: Algorithm selection based on graph size

8.2. Key Conclusions

1. All bugs successfully identified and fixed
2. All implementations now produce correct results
3. Web interface provides effective testing platform
4. Performance improves with larger graphs
5. 10 semantic commits successfully pushed to GitHub

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Appendix - System Specifications

- **OS:** Windows 10/11
- **Python:** 3.13.12
- **CPU Cores:** 8
- **RAM:** 15.94 GB
- **Web Framework:** Flask 3.1.3

Report generated on May 22, 2026

Repository: <https://github.com/sahan411/HPC-Bellman-Ford-Shortest-Path-Algorithm>

Branch: diac